

HAT-P-12b

R-1380

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_210424 · created 2026-08-01T01:49:59+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459328.7427 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.015
Transit depth (Rp/Rs)^2	1.96 +/- 0.41 [%]
Orbital Inclination (inc)	89.91 +/- 0.74
Airmass coefficient 1 (a1)	0.94 +/- 0.018
Airmass coefficient 2 (a2)	0.05 +/- 0.024
Scatter in the residuals of the lightcurve fit is	1.9 %
Best Comparison Star	#1 - [205, 271]
Optimal Method	PSF photometry
Transit Duration (day)	0.0983 +/- 0.0018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.14 ± 0.015 vs 0.1406 (deviation 0.04σ)
Duration fit vs published	0.0983 d vs 0.0975 d (deviation 0.44σ)
Mid-time O-C	-8.6 min
Depth SNR	9.3
Residual scatter	1.9 %

Light curve

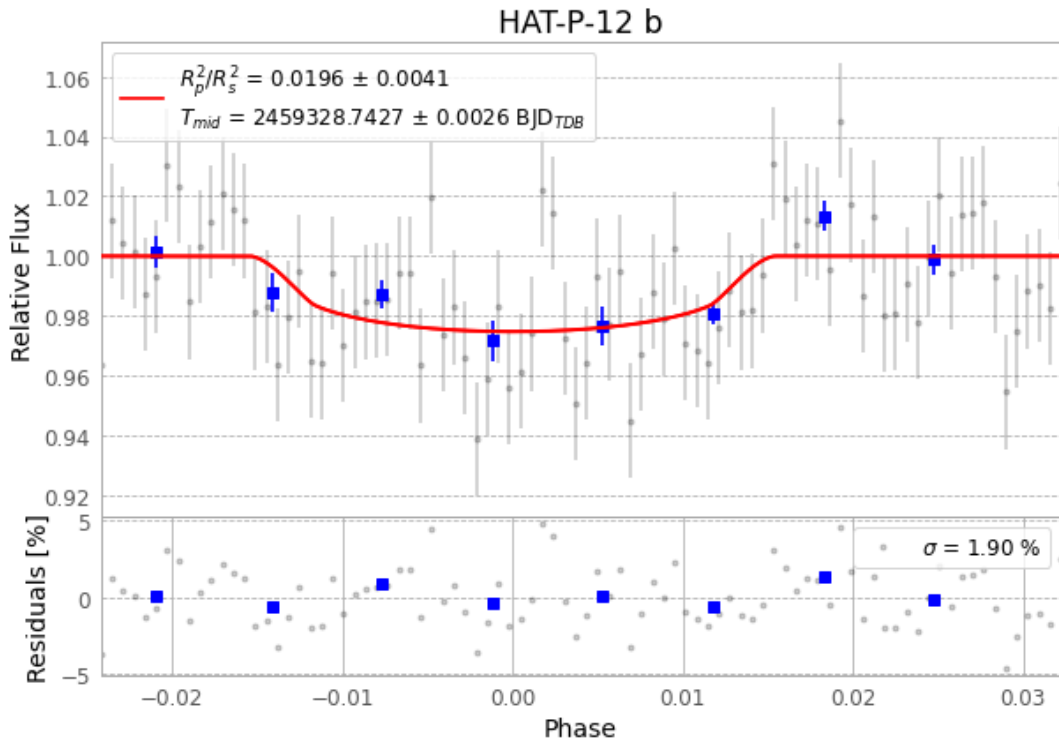


Figure 1 — FinalLightCurve_HAT-P-12 b_2021-04-23.png (SHA-256 a63e20a42a85e640...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [354, 232], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 585fff9c6fb61f44...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

88 FITS frames (58 MB), HATP-12210424035114.FITS → HATP-12210424081212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-210424.log (fd39e916f5cd...), FinalLightCurve_HAT-P-12 b_2021-04-23.png (a63e20a42a85...), FinalLightCurve_HAT-P-12 b_2021-04-23.csv (c085225cb2b0...)

Compute resources

Wall time 130.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 01:50Z.